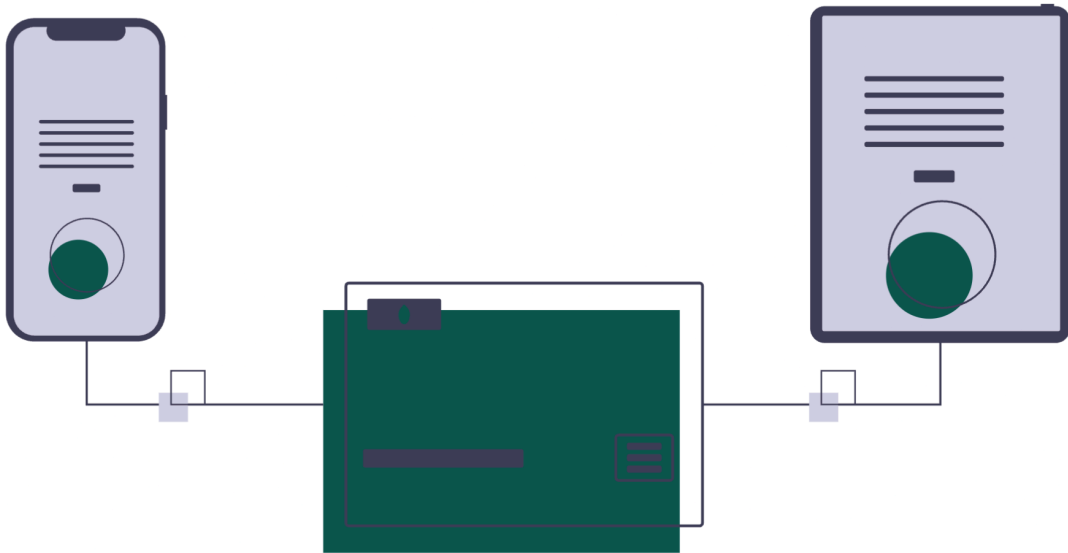


Assignment-4 of CSE405

Computer Networks



Submitted By:

Aklhak Hossain

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<https://akhlak.com>

Submitted To:

Md. Khalid Mahbub Khan

Senior Lecturer

Department of CSE

East West University

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Diagram:

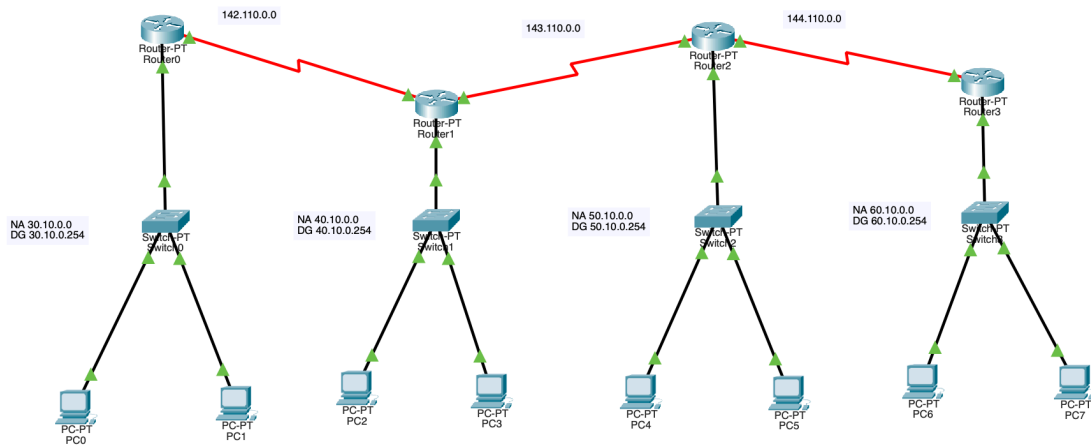


Figure: Network Topology

Router Configuration:

In the network topology above I have used the 7 Network & 8 Host Ip Addresses as shown below:

Network IP	Host IP
30.10.0.0	30.10.0.1
40.10.0.0	30.10.0.2
50.10.0.0	40.10.0.1
60.10.0.0	40.10.0.2
142.110.0.0	50.10.0.1
143.110.0.0	50.0.0.2
144.110.0.0	60.10.0.1
-	60.10.0.2

Table : Network & Host IP Addresses

Routing Protocols Overview:

Routers use dynamic routing protocols to discover networks while sharing routing updates and finding the best paths to destinations without needing manual static route entries for all subnets. The network can automatically reroute traffic because of its ability to adapt when a link goes down or when new network components are introduced.

OSPF (Open Shortest Path First):

The OSPF protocol functions as a strong and popular link-state routing protocol. The system requires all routers to create an entire network topological map which operates as the Shortest Path Tree. The system uses Dijkstra's algorithm to determine the shortest path to every destination point. The system uses a hierarchical structure which creates "Areas" to minimize routing updates and decrease memory requirements. The laboratory uses OSPF Process ID 1 as the primary system and all networks will be assigned to Area 1. The system uses wildcard masks which function as subnet mask inverses to define the specific interfaces that will take part in the routing procedure.

EIGRP (Enhanced Interior Gateway Routing Protocol):

EIGRP is a Cisco-owned advanced distance-vector routing protocol designed for efficient operation because it combines easy setup of distance-vector protocols with fast network recovery and link-state protocol performance.

1. EIGRP operates through its ability to establish neighbor relationships while it transmits only limited network updates which help to save network capacity.
2. The lab setups use an Autonomous System (AS) number of 1 for their network configurations. EIGRP uses wildcard masks in its network statements to define which interfaces will join the OSPF network just like OSPF does.

CLI Configurations:

The document presents the complete Command Line Interface (CLI) configuration details for every router in the network topology. The experiment will test OSPF and EIGRP as two different routing options. The standard production environment requires organizations to select one routing method because multiple methods lead to routing problems.

Router 0:

```
enable
config
interface fa0/0
ip address 30.10.0.254 255.0.0.0
```

```
no shut
do wr
exit
interface se2/0
ip address 142.110.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
```

OSPF 1:

```
router ospf 1
network 30.10.0.0 0.255.255.255 area 1
network 142.110.0.0 0.0.255.255 area 1
exit
```

EIGRP 1:

```
router eigrp 1
network 30.10.0.0 0.255.255.255
network 142.110.0.0 0.0.255.255
exit
```

Router 1:

```
enable
config
interface fa0/0
ip address 40.10.0.254 255.0.0.0
no shut
do wr
exit

interface se2/0
ip address 143.110.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit

interface se3/0
ip address 142.110.0.2 255.255.0.0
no shut
```

```
do wr
exit
```

OSPF 2:

```
router ospf 1
network 40.10.0.0 0.255.255.255 area 1
network 142.110.0.0 0.0.255.255 area 1
network 143.110.0.0 0.0.255.255 area 1
exit
```

EIGRP 2:

```
router eigrp 1
network 40.10.0.0 0.255.255.255
network 142.110.0.0 0.0.255.255
network 143.110.0.0 0.0.255.255
exit
```

Router 2:

```
enable
config
interface fa0/0
ip address 50.10.0.254 255.0.0.0
no shut
do wr
exit
```

```
interface se2/0
ip address 144.110.0.1 255.255.0.0
clock rate 64000
no shut
do wr
exit
```

```
interface se3/0
ip address 143.110.0.2 255.255.0.0
no shut
do wr
exit
```

OSPF 3:

```
router ospf 1
network 50.10.0.0 0.255.255.255 area 1
network 143.110.0.0 0.0.255.255 area 1
network 144.110.0.0 0.0.255.255 area 1
exit
```

EIGRP 3

```
router eigrp 1
network 50.10.0.0 0.255.255.255
network 143.110.0.0 0.0.255.255
network 144.110.0.0 0.0.255.255
exit
```

Router 3:

```
enable
config
interface fa0/0
ip address 60.10.0.254 255.0.0.0
no shut
do wr
exit

interface se3/0
ip address 144.110.0.2 255.255.0.0
no shut
do wr
exit
```

OSPF 4:

```
router ospf 1
network 60.10.0.0 0.255.255.255 area 1
network 144.110.0.0 0.0.255.255 area 1
exit
```

EIGRP 4

```
router eigrp 1
network 60.10.0.0 0.255.255.255
```

```
network 144.110.0.0 0.0.255.255
exit
```

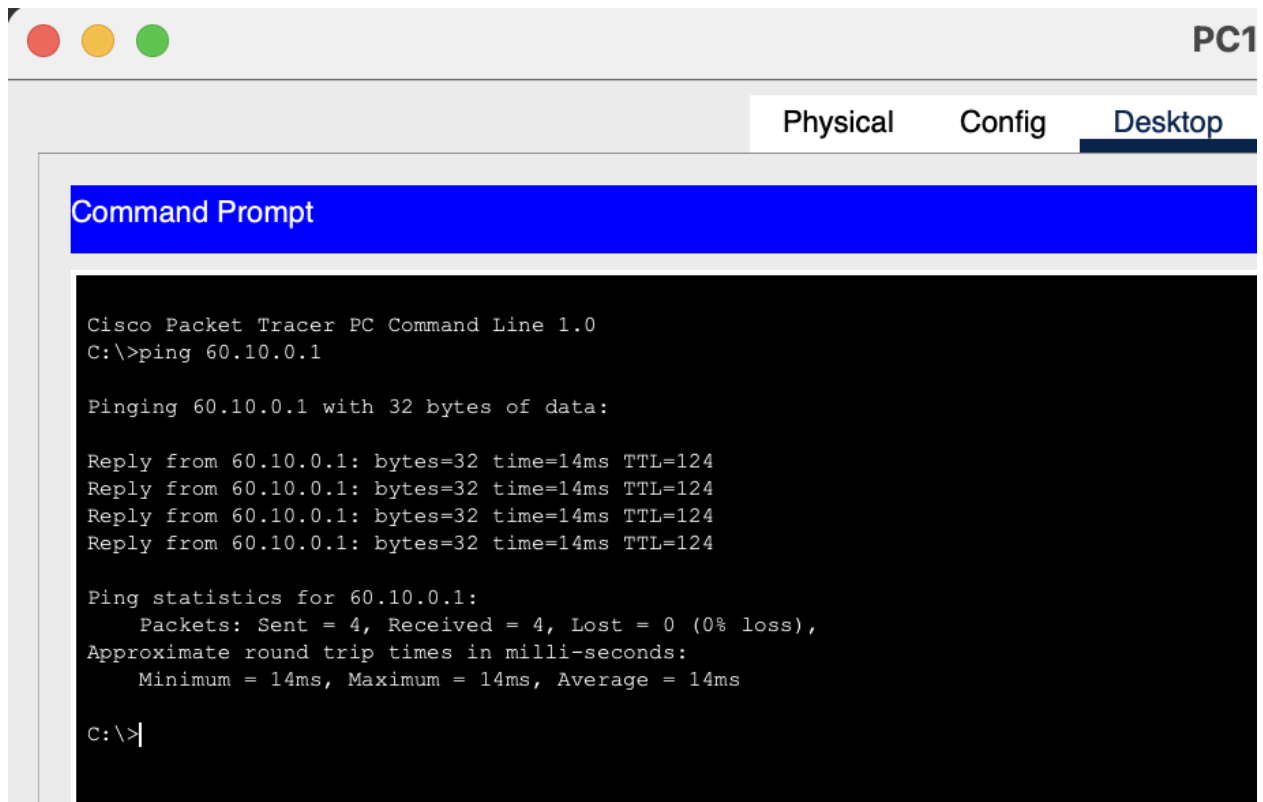
Experimental Results:

1. PDU Operations:

All networks are ping successfully.

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	PC6	PC0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC5	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC7	PC1	ICMP		0.000	N	2	(edit)	(delete)

2. Ping Operations:



The screenshot shows a Cisco Packet Tracer interface for PC1. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to 60.10.0.1, resulting in four successful replies with 14ms round-trip times and 124 TTL. The ping statistics confirm 0% packet loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 60.10.0.1

Pinging 60.10.0.1 with 32 bytes of data:

Reply from 60.10.0.1: bytes=32 time=14ms TTL=124
Reply from 60.10.0.1: bytes=32 time=14ms TTL=124
Reply from 60.10.0.1: bytes=32 time=14ms TTL=124
Reply from 60.10.0.1: bytes=32 time=14ms TTL=124

Ping statistics for 60.10.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 14ms, Average = 14ms

C:\>|
```

The ability to be able to successfully communicate with the target host indicates correct network functioning at the two ends.